

MINERAL DETECTION & MINING INTELLIGENCE SYSTEM

Software Requirements Specification (SRS)

Version 2.0 | June 2026 | Confidential

Document	Software Requirements Specification (SRS)
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0. How to Read This Document

This document defines everything MDMIS must do: every feature, processing rule, data requirement, security constraint, and system behavior. It is written so that engineers who have never visited a mine can build the system correctly, and so that mining professionals who have never read a software document can verify that every real-world need has been captured.

Every mining and technical term is explained in plain English in the Glossary (Section 1.4). Engineers should refer to it whenever they encounter an unfamiliar term.

PHASES	Requirements are tagged with the phase they belong to. Phase 1 is the MVP — the minimum system that can be deployed to a real mine site. Phases 2–5 build on it. Build Phase 1 first.
REQ CODES	Every requirement has a unique code (e.g. REQ-ING-001) so that test cases, tickets, and design documents can reference it precisely. When writing a ticket or test, always reference the code.
DECISION NEEDED	Some open questions require a client or business decision before engineering can proceed. These are listed in Section 12. Engineering should not guess — flag the item and wait for sign-off.

Phased Delivery Overview

Phase	Timeline	Theme	Key Deliverable
Phase 1 — Foundation	Weeks 1–8	Working data pipeline + auth	Sensor ingestion, RBAC, basic map display
Phase 2 — Intelligence	Weeks 9–16	AI classification	Mineral maps with confidence scores
Phase 3 — Operations	Weeks 17–24	Mining management	Traceability, grading, inventory, lab
Phase 4 — Compliance	Weeks 25–30	Regulatory readiness	OECD reports, RMB integration, audit trail
Phase 5 — Advanced	Weeks 31–40	Full production system	3D viz, live pricing, offline mobile, ERP sync

1. What Is MDMIS?

MDMIS is a software platform that replaces the fragmented, manual processes used in today's artisanal and small-scale mining operations with a single unified intelligence system. When a mining company wants to explore a land parcel, MDMIS processes the sensor data from their drones and ground equipment, tells them exactly what minerals are likely underground and at what grade, tracks those minerals from extraction through to export, and automatically generates the government compliance reports required along the way.

**SIMPLE
VERSION**

Think of MDMIS like a smart assistant for mining: it sees what is in the ground using AI, tells you whether it is worth digging, tracks everything once you dig it up, and handles all the paperwork for the government.

1.1 The Three Portals

MDMIS is delivered through three interfaces. All three share one database, but each user only sees data they are authorised to see.

Portal	Who Uses It	Primary Purpose
Field App (Mobile)	Field Operators, Drone Operators	Collect and upload sensor data; scan mineral batches; log safety events — works offline in remote areas
Web Portal	Geologists, Mine Managers, Exporters, Government Auditors	View mineral maps; manage sites; run reports; review traceability records; approve grading
Admin Portal	MDMIS System Administrators, Client Company Admins	Configure the system; manage users and organisations; view platform-wide analytics; manage integrations

1.2 User Types Supported

- Mining companies (ASM cooperatives, SME mining operations)
- Mine owners and concession holders
- Geological survey firms performing contract surveys
- Government agencies: Rwanda Mines Board (RMB), OGMR (DRC)
- Mineral exporters and trading houses requiring OECD due diligence
- Investors and financial institutions needing reserve assessments
- Safety officers and environmental compliance managers

1.3 Out of Scope

- Physical hardware: drone manufacturing, sensor production, or hardware certification
- Automation of physical mining equipment or autonomous drilling
- Airline or aviation logistics (separate platform)
- Pilot payroll, crew HR, or general ERP functions
- Deployment infrastructure decisions (cloud provider is a separate decision)

1.4 Mining & Technical Terms Glossary

Every term used in this document is defined below in plain English. Engineers unfamiliar with mining should read this section before starting work.

Term	Plain-English Explanation
MDMIS	Mineral Detection & Mining Intelligence System — the platform described in this document
Hyperspectral Imaging	A camera that captures 100+ wavelength bands (far beyond what the human eye sees). Different minerals reflect light differently, so the camera can fingerprint what mineral is on the surface.
GeoTIFF	A standard image file format for satellite and drone imagery that contains GPS coordinates — so every pixel can be placed on a real map.
HDF5	A file format for storing large scientific datasets. Used by hyperspectral cameras to store the hundreds of wavelength bands captured per image.
GPR (Ground Penetrating Radar)	A sensor that sends radar pulses into the ground and reads the echoes. Different materials (rock, mineral veins, voids) reflect differently. Works like an underground X-ray up to 10m depth.
EM (Electromagnetic Sensor)	Measures how well the soil and rock conduct electricity. Conductive zones often indicate metallic minerals like copper, gold, and cobalt.
Magnetometer	Measures variations in the Earth's magnetic field. Iron-bearing minerals like magnetite or ilmenite create detectable magnetic anomalies.
Gamma-Ray Spectrometer	Detects natural radioactivity in rock — useful for identifying uranium, thorium, and potassium concentrations near the surface.
Sentinel-2	A European Space Agency satellite that captures 13 spectral bands every 5 days globally. Free to access. Used for wide-area screening before deploying ground sensors.
Spectral Angle Mapper (SAM)	An algorithm that compares the spectral signature of a pixel to known mineral signatures in a reference library. Like matching a fingerprint.
PCA (Principal Component Analysis)	A mathematical technique that reduces the 100+ bands in hyperspectral data to a smaller set of components that capture the most important variation — used to speed up classification.
Ground Truth	Physical rock or soil samples collected at known GPS coordinates, sent to a lab for chemical analysis. Used to validate and train the AI model.
Confidence Score	A percentage (0–100%) that indicates how certain the AI model is about a mineral classification. Low scores flag zones for human review.
3TG Minerals	Tin (Cassiterite), Tantalum (Coltan), Tungsten (Wolframite), Gold — the four conflict minerals regulated under OECD and Dodd-Frank rules.
OECD Due Diligence	The Organisation for Economic Co-operation and Development's standard framework requiring mining companies to prove their minerals do not finance armed groups. Required for exports to most international buyers.
Chain of Custody	The complete documented trail showing who had possession of a mineral batch at every point from extraction to export. Required for OECD compliance.
RMB	Rwanda Mines Board — the national regulatory body that issues mining licenses, conducts site inspections, and receives compliance reports.
OGMR	Office de Géologie et des Mines du Rwanda — the geological survey arm. Also refers to the DRC equivalent in context.
AOC (Area of Concession)	A legally defined land area granted by government to a mining company for exploration or extraction.
ASM	Artisanal and Small-Scale Mining — informal or cooperative mining operations, very common in Eastern DRC and Rwanda.
RBAC	Role-Based Access Control — a security model where each user is assigned a role, and the role determines exactly what data and actions they can access.

Multi-Tenant	The system hosts multiple client organisations, each completely isolated from the others. Company A cannot see Company B's data under any circumstances.
Tenant	One client organisation and all their users, sites, data, and documents.
JWT	JSON Web Token — a secure digital key proving a user is logged in. Expires after 15 minutes for security.
MFA / TOTP	Multi-Factor Authentication — a second login check using a 6-digit code from an authenticator app (Google Authenticator, Authy).
PostGIS	A PostgreSQL extension that adds geospatial data types and queries — essential for storing and querying GPS coordinates, polygons, and map layers.
GeoJSON	A standard format for encoding geospatial data (points, polygons, lines) in JSON. Used for transferring map data between services.
Shapefile	A legacy GIS file format (.shp) for vector data (points, lines, polygons). Still widely used by ArcGIS and QGIS and commonly provided by government agencies.
WMS / WFS	Web Map Service / Web Feature Service — standard protocols for serving map layers over the internet, compatible with GIS software like QGIS and ArcGIS.
Celery	A background job system: runs slow tasks (processing large sensor files, sending emails, generating reports) without blocking the user's screen.
Redis	A fast in-memory database used for caching frequently accessed data and queuing background jobs.
Presigned URL	A temporary secure link (e.g. valid for 15 minutes) to access a private file in cloud storage. Files are never publicly accessible.
SEG-Y	A file format used to store seismic and GPR data — the industry standard for geophysical sensors.
Coltan	Columbite-tantalite — a mineral ore containing tantalum and niobium, used in smartphone capacitors and aerospace components. The primary high-value mineral in the Rubaya area near Goma.
Cassiterite	The primary ore of tin. Major export commodity from the Masisi Territory and wider Kivu region.
Wolframite	The primary ore of tungsten. Used in steel hardening and aerospace manufacturing. Found throughout the Kivu region.

2. System Architecture

FLOW	<i>Sensor data collected in the field → uploaded to MDMIS → pre-processed and fused → classified by AI → mineral maps generated → operations tracked → compliance reports auto-generated → exported to regulator. All roles share one database, but each organisation only sees their own data.</i>
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2.1 Technology Stack

Layer	Technology	Why We Use It
Backend API	Python (FastAPI)	Best ecosystem for geospatial and ML libraries; async-native for handling large file uploads without blocking; faster than Django REST Framework for I/O-heavy workloads
Geospatial Processing	GDAL, Rasterio, GeoPandas, Fiona	Industry standard for GIS raster and vector data processing; GDAL reads every satellite and drone format; PostGIS queries run via GeoPandas
Hyperspectral Analysis	Spectral (Python), PySpTools	Purpose-built libraries for mineral spectral signature matching; implement SAM, PCA unmixing, and wavelength band ratio calculations
ML / Classification	scikit-learn, PyTorch	scikit-learn for Random Forest and SVM classifiers; PyTorch for CNN-based image classification on hyperspectral cubes
Database	PostgreSQL 16 + PostGIS 3.x	Reliable, ACID-compliant, row-level security; PostGIS adds native geospatial queries (intersect, contain, distance) on scan polygons and GPS points
File Storage	S3-compatible object store (MinIO / AWS S3)	Stores large GeoTIFF, HDF5, SEG-Y sensor files and generated PDFs; pluggable to any provider via env variable
Frontend	Next.js 14 + React 18	Server-side rendering for fast initial load; App Router for complex multi-page portal; used for Web Portal and Admin Portal
Map Rendering	MapLibre GL JS + Mapbox tiles	Interactive WebGL mineral maps with layer controls; vector tile rendering; supports offline tile caching for field use
Mobile / Field App	React Native + Expo	Single codebase for iOS and Android; offline-capable with SQLite local storage; syncs on reconnection
Background Jobs	Celery + Redis	Handles slow tasks (image processing, report generation, email) without blocking user screens; job queue depth monitored in admin
ML Training Infrastructure	GPU cloud instance (AWS g4dn / GCP T4)	Hyperspectral classification is compute-heavy; training jobs run on GPU; inference runs on standard CPU instances
Containerisation	Docker + Docker Compose	Packages all services into containers deployable on any cloud or on-premise server without environment differences
CI/CD	GitHub Actions	Auto-tests and deploys code on every merge to main; test suite must pass before any deployment
Error Tracking	Sentry	Alerts the team when any component crashes in production; captures stack traces and affected user context

Email	SendGrid	Transactional email delivery for notifications, alerts, and compliance report distribution
SMS	Africa's Talking (primary) / Twilio (fallback)	SMS alerts for critical safety events and system notifications; Africa's Talking has better coverage and pricing across East Africa

2.2 Multi-Tenant Data Isolation

Every client organisation is a separate tenant. All database tables storing client data include an `org_id` column. Every query **MUST** filter by `org_id` at the ORM layer — enforced in application code, not just in the database. Organisation A can never see Organisation B's scan data, mineral maps, or compliance records under any circumstances.

2.3 The Nine-Step Data Pipeline

This is the core technical process that MDMIS executes every time sensor data arrives. Engineers must understand this flow before building any module.

Step	What Happens
01 — Data Acquisition	Geological data collected from drone hyperspectral cameras, GPR, EM sensors, magnetometers, gamma-ray spectrometers, satellite imagery, and laboratory samples. Uploaded via Field App (offline sync) or direct API push.
02 — Data Preprocessing	Validation, noise removal, atmospheric correction, signal normalisation, and geospatial alignment. Malformed or incomplete files are rejected with a detailed error report.
03 — Multi-Sensor Fusion	Data from all sensor sources is co-registered to a common spatial grid and merged into a unified geological dataset. Fusion improves classification accuracy by combining surface (hyperspectral) with subsurface (GPR/EM) information.
04 — AI Mineral Classification	Machine learning algorithms analyse the fused dataset: identify mineral types, estimate concentration, assign confidence scores (0–100%), and flag anomalies that do not match known signatures.
05 — Geological Mapping	Production of georeferenced mineral distribution maps: surface heat maps from hyperspectral, subsurface cross-sections from GPR, resource volume calculations per zone.
06 — Resource Estimation	Quantitative reserve analysis combining mineral type, estimated depth, spatial extent, and grade to calculate estimated tonnage and in-situ value per zone.
07 — Risk & Safety Analysis	Evaluation of geological instability indicators, proximity to fault lines and water tables, gas risk indicators, and historical incident data. Risk score assigned per zone.
08 — Traceability & Compliance	Every mineral batch tracked from extraction through storage, transportation, and export. Unique chain-of-custody ID generated at scan time. OECD and RMB reports generated automatically.
09 — Reporting & Decision Support	Mineral intelligence dashboards, resource estimation reports, safety assessments, compliance documentation, grading certificates, and investment reports made available to authorised users.

3. User Roles & Access Control

3.1 Role Matrix

The table below defines every user role, which portal they access, and what they are permitted to do. No role may exceed its permissions — enforcement is at the API middleware layer.

Role	Portal	Permissions
Field Operator	Field App	Upload sensor data; scan and tag mineral batches with QR codes; log safety incidents; view assigned site boundaries; works offline
Drone Operator	Field App	Upload flight logs and hyperspectral GeoTIFF/HDF5 files; view upload status; read-only on site details
Geologist	Web Portal	View and annotate mineral maps; review classification results; validate ground truth samples; generate geological reports; manage lab sample records
Mine Manager	Web Portal	All geologist permissions + manage site operations; view production and inventory ledger; approve grading certificates; review safety risk scores; view compliance status
Safety Officer	Web Portal	View real-time safety risk scores; acknowledge and log safety incidents; receive and act on safety alerts; read-only on all geological data
Compliance / Export Manager	Web Portal	View traceability chain-of-custody records; generate and download OECD due diligence reports; manage export documentation; read-only on operational data
Government Auditor (RMB / OGMR)	Web Portal	Read-only access scoped to licenced concessions they regulate; view compliance records, grading certificates, traceability reports; cannot edit any data
Investor / Read-Only	Web Portal	View resource estimates, production summaries, and financial analytics for authorised sites only; no operational data
Client Company Admin	Web Portal + Admin Portal	All client permissions + manage users, sites, aircraft records; configure notification preferences; view organisation-wide audit trail
MDMIS System Admin	Admin Portal	Full platform access: manage all organisations; configure integrations; view system health; suspend accounts; access immutable audit log; manage ML model versions

3.2 Authentication Requirements

REQ-ACC-001 [Phase 1]	All users MUST log in with email and password. Passwords MUST be hashed using bcrypt (work factor ≥ 12). Plain-text passwords must NEVER be stored in the database or logs.
REQ-ACC-002 [Phase 1]	TOTP Multi-Factor Authentication (Google Authenticator compatible) MUST be required for Client Company Admins, Compliance Managers, Government Auditors, and all MDMIS internal staff. MFA must be configured before accessing any protected screen.
REQ-ACC-003 [Phase 1]	If a user enters the wrong password 5 times within 15 minutes, their account MUST be locked automatically. An unlock email (valid 1 hour, single-use) is sent. The lockout event MUST be logged with the IP address and timestamp.

REQ-ACC-004 [Phase 1]	Sessions MUST use JWT access tokens (15-minute expiry) plus an httpOnly refresh token (30-day expiry). A new refresh token MUST be issued on every refresh (token rotation). Expired or invalid tokens return HTTP 401.
REQ-ACC-005 [Phase 1]	Every API endpoint MUST check the user's role in middleware before running any business logic. Insufficient permissions MUST return HTTP 403 Forbidden with a machine-readable error code.
REQ-ACC-006 [Phase 1]	When a user sets a new password, the system MUST check it against HaveIBeenPwned using the k-Anonymity method (only the first 5 characters of the SHA-1 hash are sent). Compromised passwords MUST be rejected with a clear message.
REQ-ACC-007 [Phase 1]	Changing a password MUST revoke all existing refresh tokens for that user. All other active sessions are invalidated immediately.
REQ-ACC-008 [Phase 1]	Password minimum requirements: 12 characters, at least one uppercase, one number, one special character. Admin accounts must rotate passwords every 90 days.
REQ-ACC-009 [Phase 2]	Field App users operating in offline mode MUST authenticate with a locally cached credential that expires after 72 hours of no server sync. Expired offline sessions require network reconnection to renew.

4. Core Module Functional Requirements

This section defines the detailed requirements for each functional module. Every requirement has a unique code and a phase tag. Engineers building a phase should implement all requirements tagged for that phase.

4.1 Sensor Data Ingestion Module

This module is the entry point for all raw data. Without it, nothing else in the system can function. It must be rock-solid, well-validated, and able to handle the largest sensor files without crashing or timing out.

WHY IT MATTERS	<i>A corrupted or misaligned sensor file will silently produce wrong mineral maps downstream. Strict validation here prevents incorrect data from ever reaching the AI classifier.</i>
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4.1.1 Accepted Sensor Formats

Sensor	Data Format	Coverage Depth	Primary Use
Drone Hyperspectral Camera	HDF5, GeoTIFF (100+ bands)	Surface (top ~5 cm)	Mineral spectral fingerprinting
Ground Penetrating Radar (GPR)	Binary / SEG-Y	Up to 10m underground	Subsurface mineral boundary detection
Electromagnetic (EM) Sensor	CSV / Binary	2–50m depending on frequency	Rock conductivity mapping
Magnetometer	CSV / ASCII	Shallow to deep depending on target	Iron-bearing mineral detection
Gamma-Ray Spectrometer	ASCII / CSV	Near-surface (<1m)	Radioactive element detection (U, Th, K)
Sentinel-2 Satellite	GeoTIFF (13 bands)	Surface only	Large-area screening before ground deployment
Lab Spectrometer	CSV / JSON	N/A (ground truth)	AI model training and validation

4.1.2 Ingestion Requirements

REQ-ING-001 [Phase 1]	The system MUST parse and validate all sensor formats listed in Section 4.1.1. Validation checks: file format integrity, geospatial metadata presence (bounding box), sensor type identifier, minimum file size (reject obviously empty files).
REQ-ING-002 [Phase 1]	The system MUST reject malformed or unreadable files and return a detailed error report to the uploader specifying: which validation check failed, the file name, and a suggested corrective action.
REQ-ING-003 [Phase 1]	The system MUST support both manual upload via the web UI and automated transfer via API push. Maximum file size per upload: 10 GB. Resumable uploads (multipart) MUST be supported for files over 100 MB.
REQ-ING-004 [Phase 1]	The system MUST log every ingestion event with: ingestion timestamp (UTC), sensor type, operator user ID, GPS bounding box, file size, and upload method (manual / API). This log is immutable.

REQ-ING-005 [Phase 1]	A 1 GB GeoTIFF file MUST be pre-processed and ready for classification within 5 minutes on standard cloud compute (4 vCPU, 16 GB RAM). Processing MUST run as a background Celery task — it MUST NOT block the user's upload confirmation screen.
REQ-ING-006 [Phase 2]	The Field App MUST support offline data collection: sensor files are queued locally in the device's storage. When connectivity is restored, the queue MUST sync automatically within 2 minutes. Upload status must be visible in the Field App during sync.
REQ-ING-007 [Phase 1]	All uploaded sensor files MUST be stored in organisation-specific storage paths. No file from Organisation A is ever accessible under Organisation B's path. Access to stored files MUST use presigned URLs valid for 15 minutes maximum.
REQ-ING-008 [Phase 1]	All uploaded files MUST be malware-scanned (ClamAV self-hosted sidecar) before being made available for processing. Files that fail the malware scan MUST be quarantined and the uploading user notified.

4.2 Geological Mapping & Resource Estimation

This module transforms processed sensor data into the visual mineral maps that geologists and mine managers use to make extraction decisions. Maps must be accurate, layerable, and easy to interpret by both technical and non-technical users.

REQ-GEO-001 [Phase 1]	The system MUST generate georeferenced mineral distribution maps from processed sensor data. Maps MUST display: mineral type, estimated depth, spatial extent (polygon), and confidence score per zone. Every map pixel must be traceable to the sensor scan that produced it.
REQ-GEO-002 [Phase 1]	The system MUST support polygon-based land parcel definition via: (a) manual GPS coordinate entry, (b) shapefile import (.shp), or (c) on-screen drawing on the map interface.
REQ-GEO-003 [Phase 2]	The system MUST support importing historical geological survey data (existing DRC/Rwanda government records in shapefile or PDF format) and displaying them as a reference layer alongside new scan data.
REQ-GEO-004 [Phase 1]	The system MUST support multi-layer map views: users can toggle layers on and off including surface hyperspectral, subsurface GPR cross-section, satellite base map, concession boundaries, and safety risk overlay.
REQ-GEO-005 [Phase 2]	The system MUST calculate estimated resource tonnage per mineral zone using standard volumetric methods (zone area × estimated depth × estimated density). The calculation methodology MUST be documented in the output report.
REQ-GEO-006 [Phase 1]	All map views MUST load within 3 seconds on a standard broadband connection (20 Mbps). Vector tile rendering is preferred over raster tiles for performance.
REQ-GEO-007 [Phase 2]	Users MUST be able to compare any two historical versions of a mineral map for the same site on a split-screen view.
REQ-GEO-008 [Phase 5]	The system MUST support interactive 3D subsurface visualisation combining GPR depth profiles and EM conductivity data in a WebGL viewer. Users can rotate, zoom, and take cross-section slices.

4.3 ML Mineral Classification Engine

This is the AI brain of the platform. It takes the pre-processed sensor data and outputs a classification: what mineral is likely here, how confident we are, and what the grade appears to be. Every decision it makes must be explainable and traceable.

KEY RULE	<i>The AI never makes a silent decision. Every output must include a confidence score, the model version that produced it, and the input data it was based on. If confidence is below threshold, the zone is flagged for human review — the system does not guess.</i>
REQ-ML-001 [Phase 2]	The system MUST classify mineral types from hyperspectral and fused sensor data. Minimum target accuracy at Phase 2 launch: 85% on the validation dataset (which must contain at least 200 lab-confirmed ground truth samples).
REQ-ML-002 [Phase 2]	Every classification output MUST include: (a) mineral type prediction, (b) confidence score 0–100%, (c) model version ID used, (d) the training dataset version the model was trained on, (e) timestamp of classification.
REQ-ML-003 [Phase 2]	The system MUST support a configurable confidence threshold per organisation (default: 70%). Zones where confidence falls below the threshold MUST be displayed as "Unconfirmed — Lab Sample Required" on the map.
REQ-ML-004 [Phase 3]	The system MUST support model retraining using new lab-validated ground truth samples. Retraining MUST run as a background job on the GPU training instance. A new model version is created; the previous version is archived, not deleted.
REQ-ML-005 [Phase 2]	All model versions MUST be stored with: version number, training dataset reference, accuracy metrics (precision, recall, F1 per mineral class), training date, and deployed-by user ID. No version may be deleted — only deprecated.
REQ-ML-006 [Phase 2]	The system MUST apply spectral band ratios as preprocessing features before classification. Band ratios to implement as a minimum: Iron Oxide (B11/B8), Carbonate (B11/B12), Clay (B11/B8A) — all based on Sentinel-2 band numbering.
REQ-ML-007 [Phase 2]	The system MUST detect and flag spectral anomalies that do not match any known mineral signature in the reference library. Flagged anomalies appear on the map with an "Unknown Signature" label and are queued for geologist review.
REQ-ML-008 [Phase 2]	Classification inference (return mineral classification for a single scanned zone) MUST complete within 30 seconds on standard CPU compute after preprocessing is complete.

4.4 Mineral Traceability Module

This module is MDMIS's most critical compliance feature. It creates the chain of custody that proves to international buyers that minerals were legally and responsibly sourced. Without it, no OECD-compliant exporter can use the system.

4.4.1 Batch Status Values

Every mineral batch moves through the following statuses. Engineers must implement exactly these values — no custom statuses.

Status	What It Means
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Scanned	Batch detected and tagged by Field Operator at the mine face. Chain-of-custody ID generated.
Weighed	Batch weighed and volume recorded. Quantity officially registered in inventory.
Graded	Mineral grade assessed (detected or lab-confirmed). Grading certificate attached.
In Storage	Batch moved to secure storage at mine site. Location and quantity recorded.
In Transit	Batch dispatched from mine site to processing facility or export point. Carrier and route logged.
At Processing	Batch received at processing or beneficiation facility.
Exported	Batch cleared customs and formally exported. Export documentation attached.
Rejected	Batch rejected at any stage (failed inspection, compliance flag). Reason recorded.

REQ-TRACE-001 [Phase 3]	Every mineral batch MUST be assigned a unique, system-generated chain-of-custody ID at the moment of first scan by a Field Operator. The ID format: COC-{SITE_CODE}-{YYYYMMDD}-{SEQUENCE}. Once assigned, this ID is permanent and immutable.
REQ-TRACE-002 [Phase 3]	The system MUST record the following data points at every custody transfer event: (a) batch chain-of-custody ID, (b) from-party and to-party, (c) GPS coordinates of transfer point, (d) timestamp (UTC), (e) quantity transferred, (f) authorising user ID, (g) digital signature.
REQ-TRACE-003 [Phase 3]	Traceability records are IMMUTABLE . No deletion is permitted at any stage. Corrections MUST be made as new amendment records with full audit trail: original value, corrected value, reason, user ID, timestamp.
REQ-TRACE-004 [Phase 4]	The system MUST generate OECD-compliant due diligence reports for 3TG minerals (Tin, Tantalum, Tungsten, Gold). Reports MUST be exportable as PDF and XML. Report template must match the OECD Annex II Due Diligence Guidance format.
REQ-TRACE-005 [Phase 3]	The system MUST support QR code generation per mineral batch for physical tagging at the mine site. Scanning the QR code with the Field App opens the full chain-of-custody record for that batch.
REQ-TRACE-006 [Phase 4]	The system MUST flag any discrepancy where the quantity at any transfer point does not match the recorded inventory quantity for that batch — within a tolerance of $\pm 2\%$. Flagged discrepancies are routed to the Mine Manager and Compliance Manager.

4.5 Mineral Grading & Value Estimation

REQ-GRADE-001 [Phase 3]	The system MUST classify mineral purity and grade from spectral and lab data per the following grading standards: Coltan (% Ta ₂ O ₅), Gold (fineness parts per 1000), Cassiterite (% Sn), Wolframite (% WO ₃). Grade values must be stated as "Detected" or "Lab-Confirmed" — these must never be mixed without clear labelling.
REQ-GRADE-002 [Phase 3]	The system MUST integrate live commodity price feeds for Coltan, Gold, Cassiterite, and Wolframite for real-time value estimation per batch. Price source and timestamp must be displayed alongside every value estimate. [See Section 6 for commodity price API integration.]
REQ-GRADE-003 [Phase 3]	The system MUST generate grading certificates as signed, downloadable PDFs. Each certificate MUST include: batch ID, mineral type, grade result (detected/confirmed), quantity, site of origin,

	GPS coordinates, date of assessment, grading methodology, and generating operator's digital signature.
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REQ-GRADE-004 [Phase 3]	Value estimates MUST clearly distinguish between "Detected Grade" (from AI/spectral analysis) and "Lab-Confirmed Grade" (from laboratory analysis). Detected grade estimates are advisory only and MUST include a disclaimer on all exports.
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4.6 Safety Regulations & Risk Prediction

REQ-SAFE-001 [Phase 1]	The system MUST maintain a configurable rule engine for safety threshold alerts. Configurable parameters include: gas concentration thresholds (CO, CH4, H2S), geological instability score threshold, slope angle limit, and proximity to fault lines. Thresholds are set per site by the Mine Manager.
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REQ-SAFE-002 [Phase 2]	The system MUST generate a risk score (0–100) per excavation zone based on: geological instability indicators from GPR data, proximity to water table, proximity to mapped fault lines, slope gradient, and historical incident data for that zone.
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REQ-SAFE-003 [Phase 2]	When a risk threshold is breached, the system MUST send real-time alerts to designated Safety Officers via: (a) in-app push notification, (b) SMS (Africa's Talking), (c) email. Alert MUST include: zone ID, risk score, threshold breached, and recommended action.
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REQ-SAFE-004 [Phase 1]	The system MUST log all safety incidents with: timestamp (UTC), GPS location, sensor readings at time of incident, operator ID, incident description, and acknowledgement record (who acknowledged, when). Safety incident logs are immutable.
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REQ-SAFE-005 [Phase 2]	Safety Officers MUST be able to mark a zone as "Restricted" which blocks the system from generating extraction recommendations for that zone until the restriction is lifted with a documented reason.
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4.7 Production & Inventory Management

REQ-INV-001 [Phase 3]	The system MUST track extraction volumes per mineral, per site, per time period (daily, weekly, monthly). Data source: Field Operator weight entries confirmed by Mine Manager.
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REQ-INV-002 [Phase 3]	The system MUST maintain an inventory ledger recording: extracted quantities, quantities in storage (by location), quantities in transit, quantities exported. The ledger updates in real time as batch statuses change.
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REQ-INV-003 [Phase 4]	The system MUST generate production variance reports comparing estimated yield (from resource estimation) to actual yield (from inventory ledger) per site per period.
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REQ-INV-004 [Phase 3]	The system MUST flag inventory discrepancies where total extracted quantity exceeds the licensed extraction volume for that concession and period. Flagged discrepancies are automatically reported to the Compliance Manager and can be escalated to the Government Auditor.
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4.8 Ground Truth & Lab Sample Management

REQ-LAB-001 [Phase 2]	The system MUST maintain a database of all lab-analysed soil and rock samples with: GPS coordinates, depth of collection, mineral composition (by element/compound), analysis date, lab reference number, and analyst name.
REQ-LAB-002 [Phase 2]	Lab results MUST be linkable to corresponding scan sessions for model validation. The link enables accuracy assessment: compare the AI's classification for that GPS point to the lab-confirmed result.
REQ-LAB-003 [Phase 3]	The system MUST track sample chain of custody from field collection to lab receipt to result upload: who collected it, when, GPS location, who received it at the lab, when results were uploaded.
REQ-LAB-004 [Phase 2]	The system MUST alert geologists when an area has insufficient lab sample coverage for reliable AI classification — defined as: fewer than 1 lab sample per 500m ² for high-value mineral zones.

4.9 Mine Planning & Design Support

REQ-PLAN-001 [Phase 3]	The system MUST generate excavation zone recommendations based on mineral concentration maps, resource estimates, and safety risk scores. Recommendations are ranked by estimated value-to-risk ratio.
REQ-PLAN-002 [Phase 3]	The system MUST provide an estimated extraction cost vs. mineral value analysis per recommended zone. Inputs: estimated tonnage, commodity price (live feed), estimated extraction cost per tonne (configurable by Mine Manager).
REQ-PLAN-003 [Phase 3]	The system MUST identify and visually flag environmentally or structurally sensitive zones on the mine planning map: water tables, mapped fault lines, protected forest boundaries, and populated areas.

4.10 Transportation Tracking

REQ-TRANS-001 [Phase 3]	The system MUST log mineral shipment events at each waypoint: departure site, route waypoints, destination, quantity, carrier name and licence, vehicle registration. Each waypoint entry must be confirmed by the receiving party.
REQ-TRANS-002 [Phase 3]	The system MUST flag any shipment where the declared quantity at departure does not match the quantity received at destination (beyond $\pm 2\%$ tolerance). Flagged shipments generate a discrepancy report routed to the Compliance Manager.
REQ-TRANS-003 [Phase 3]	All transportation records MUST be automatically linked to the corresponding traceability chain-of-custody record for the batches in that shipment.

4.11 Multi-Site Management

REQ-SITE-001 [Phase 1]	The system MUST support multiple land parcels (sites) per client organisation. Each site has its own: sensor data repository, mineral maps, safety configuration, inventory ledger, and access control list.
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REQ-SITE-002 [Phase 2]	Client Company Admins MUST be able to view a cross-site portfolio dashboard showing: total resource estimates across all sites, active safety alerts, pending compliance actions, and inventory summary — all in one view.
REQ-SITE-003 [Phase 4]	The system MUST support site archiving for completed surveys. Archived sites remain fully accessible for historical reference and compliance audit but do not appear in active operational views by default.

4.12 Reporting & Export

REQ-RPT-001 [Phase 3]	The system MUST generate scan reports in PDF and Excel formats containing: site info, mineral distribution map image, classification results table, confidence scores per zone, resource estimation summary, and methodology summary including model version and training dataset reference.
REQ-RPT-002 [Phase 4]	The system MUST support Rwanda Mines Board (RMB) formatted submission reports. The report template must match the format currently mandated by RMB. [Open Item OQ-07 — RMB template must be obtained before development of this requirement begins.]
REQ-RPT-003 [Phase 4]	All report exports MUST include a data lineage section: sensor sources used, dates of data collection, model version, processing steps applied. This allows regulators to verify how conclusions were reached.
REQ-RPT-004 [Phase 4]	Reports MUST support digital signature by the generating operator using their MDMIS credentials. Signed reports display the operator's name, role, and a verified timestamp.
REQ-RPT-005 [Phase 4]	A 7-year immutable audit trail of all system actions MUST be maintained: every request submitted, every report generated, every user action on sensitive data, every status change, every model version deployed. The audit trail cannot be modified or deleted by any user, including system admins.

5. Non-Functional Requirements

5.1 Performance

Metric	Requirement
Hyperspectral image processing	Process a 1 GB GeoTIFF in under 5 minutes on 4 vCPU / 16 GB RAM standard cloud instance
Classification inference	Return mineral classification for a scanned zone within 30 seconds after preprocessing completes
Map / dashboard load time	All map and dashboard views load within 3 seconds at P95 on 20 Mbps broadband
API response time	All user-facing API responses return within 500ms at P95 under normal load
Offline sync	Queued field data syncs within 2 minutes of network reconnection
Report generation	PDF report generated within 60 seconds for a single-site scan session report
Concurrent users	Phase 1: 50 concurrent; Phase 3: 500 concurrent; Phase 5: 2,000 concurrent — all without performance degradation

5.2 Security

REQ-SEC-001 [Phase 1]	All data in transit MUST be encrypted using TLS 1.3. Older TLS versions (1.0, 1.1, 1.2) MUST be disabled on all servers.
REQ-SEC-002 [Phase 1]	All data at rest MUST be encrypted using AES-256. This includes database storage, object storage, and any backup media.
REQ-SEC-003 [Phase 1]	All uploaded files MUST be stored in organisation-specific storage paths. No file is ever publicly accessible. All access MUST use presigned URLs with a maximum validity of 15 minutes.
REQ-SEC-004 [Phase 1]	All API inputs MUST be validated and sanitized. SQL injection, XSS, and CSRF protection MUST be enforced from Phase 1.
REQ-SEC-005 [Phase 1]	Rate limiting MUST be applied to all authentication endpoints: maximum 10 requests per minute per IP. Exceeding the limit returns HTTP 429 with a Retry-After header.
REQ-SEC-006 [Phase 1]	Geological scan data is commercially sensitive. Access to any scan data, mineral map, or resource estimate MUST be scoped strictly to the owning organisation. MDMIS system admins accessing a client organisation's data MUST have their access logged in the immutable audit trail.
REQ-SEC-007 [Phase 4]	An independent security penetration test MUST be completed before Phase 4 deployment. Annual penetration tests thereafter. No critical or high-severity finding may be open at time of deployment.

5.3 Reliability & Availability

REQ-REL-001 [Phase 1]	System uptime target: 99.5% (approximately 43 hours of permitted downtime per year). Planned maintenance windows must be communicated 48 hours in advance.
REQ-REL-002 [Phase 1]	All data MUST be backed up daily with point-in-time recovery for the previous 30 days. Backup restore to a clean environment must be tested monthly and complete within 4 hours.
REQ-REL-003 [Phase 1]	The system MUST degrade gracefully. If the ML classification service is unavailable, raw data ingestion, map viewing of existing data, and report access MUST remain functional. The UI must clearly indicate which capabilities are temporarily unavailable.

5.4 Scalability

REQ-SCAL-001 [Phase 1]	The processing pipeline architecture MUST support horizontal scaling. Multiple worker instances can be added behind a load balancer to handle increased concurrent ingestion jobs.
REQ-SCAL-002 [Phase 3]	The database must handle a minimum of 10 million scan records without query performance degradation. Table partitioning by org_id and scan_date MUST be implemented before Phase 3 deployment.
REQ-SCAL-003 [Phase 3]	Object storage MUST support a minimum of 50 TB of raw sensor data with archival tiering: data older than 1 year moves to lower-cost archival storage automatically.

5.5 Usability

REQ-USE-001 [Phase 1]	The Field App interface MUST be operable with minimal training. Maximum onboarding time for a Field Operator: 2 hours. Core workflows (upload sensor file, scan and tag batch, log safety event) must require no more than 4 taps each.
REQ-USE-002 [Phase 1]	All map interfaces MUST be responsive and usable on tablet devices (minimum 10-inch screen, 1280×800 resolution) used in the field.
REQ-USE-003 [Phase 1]	The system MUST display all confidence scores and uncertainty indicators clearly and prominently. There must be no silent failures — if a classification is uncertain, the user sees it.

6. External Integrations

All external API calls MUST run as background tasks via Celery. They must NOT block any user-facing HTTP response. If an external service is down, the system degrades gracefully and queues the request for retry.

Integration	What It Does	Direction	Protocol	Phase
Rwanda Mines Board (RMB)	Submit regulatory compliance reports; receive licence validation	Outbound	PDF/XML structured report	Phase 4
Commodity Price API (LME / Kitco)	Live prices: Coltan, Gold, Cassiterite, Tungsten — daily refresh for value estimates	Inbound	REST / JSON	Phase 3
Sentinel-2 (ESA Copernicus Hub)	Download satellite imagery for wide-area screening (free access)	Inbound	STAC / HTTP download	Phase 1
DJI Terra / Pix4D	Import processed drone orthomosaic and hyperspectral outputs	Inbound	GeoTIFF export / API	Phase 1
GIS Platforms (QGIS, ArcGIS)	Import/export geological layers; publish WMS/WFS map services	Bidirectional	Shapefile, WMS, WFS	Phase 2
Lab Information Systems	Receive ground truth lab analysis results per sample batch	Inbound	CSV / JSON upload	Phase 2
ERP / Inventory Systems	Sync production volumes and inventory data with client ERP	Bidirectional	REST API webhooks	Phase 5
SendGrid	All transactional emails: alerts, reports, compliance notifications	Outbound	SMTP / REST API	Phase 1
Africa's Talking	SMS alerts for safety events and critical notifications in East Africa	Outbound	REST API	Phase 1
ClamAV	Malware scan all uploaded sensor files. Self-hosted Docker sidecar.	Internal	Local socket	Phase 1
HaveIBeenPwned API	k-Anonymity password compromise check at	Outbound	REST API (HTTPS)	Phase 1

	registration and password change			
Public REST API + Webhooks	Allow client organisations to submit data and receive status updates programmatically	Bidirectional	REST / JSON + WebSocket	Phase 5

7. Key Database Models

These are the primary database tables. Every model storing client data includes an `org_id` column. All primary keys are UUIDs. All timestamps are stored in UTC.

7.1 ScanSession

Created each time a field operator uploads sensor data for a site visit.

Field	Type	Description
id	UUID (PK)	Unique scan session ID
org_id	UUID (FK)	Which client organisation
site_id	UUID (FK)	Which mine site
operator_id	UUID (FK)	Field Operator who uploaded
sensor_types	ARRAY[ENUM]	Which sensors were included: hyperspectral / gpr / em / magnetometer / gamma / satellite / lab
bounding_box	GEOMETRY (PostGIS)	GPS polygon of the survey area
status	ENUM	uploaded / preprocessing / ready / classifying / complete / failed
uploaded_at	TIMESTAMP	UTC time of upload
processed_at	TIMESTAMP (nullable)	UTC time preprocessing completed

7.2 MineralZone

One record per detected mineral zone — created by the classification engine for each distinct area identified on the map.

Field	Type	Description
id	UUID (PK)	Unique zone ID
scan_session_id	UUID (FK)	Linked scan session
org_id	UUID (FK)	Which client organisation (denormalised for query performance)
mineral_type	ENUM	coltan / gold / cassiterite / wolframite / cobalt / copper / gemstone / unknown
confidence_score	INTEGER	AI confidence 0–100
geometry	GEOMETRY (PostGIS POLYGON)	GPS polygon of the mineral zone
estimated_depth_m	DECIMAL (nullable)	Estimated depth in metres
estimated_tonnage	DECIMAL (nullable)	Estimated resource tonnage
model_version_id	UUID (FK)	Which ML model version produced this classification
status	ENUM	unconfirmed / geologist_reviewed / lab_confirmed / rejected
flagged_anomaly	BOOLEAN	True if spectral signature did not match any known mineral
created_at	TIMESTAMP	UTC time of classification

7.3 MineralBatch (Traceability)

Created by a Field Operator when physically tagging and weighing a batch of extracted mineral.

Field	Type	Description
id	UUID (PK)	Unique batch ID
coc_id	TEXT	Chain-of-custody ID: COC-{SITE_CODE}-{YYYYMMDD}-{SEQ}
org_id	UUID (FK)	Which client organisation
site_id	UUID (FK)	Site of extraction
mineral_type	ENUM	Same enum as MineralZone.mineral_type
origin_gps	GEOMETRY (PostGIS POINT)	GPS point of extraction
extraction_timestamp	TIMESTAMP	UTC time of extraction
weight_kg	DECIMAL	Weight in kilograms
grade_detected	DECIMAL (nullable)	AI-estimated grade (%)
grade_confirmed	DECIMAL (nullable)	Lab-confirmed grade (%) — null until lab result uploaded
status	ENUM	scanned / weighed / graded / in_storage / in_transit / at_processing / exported / rejected
qr_code_url	TEXT	Object storage URL of the generated QR code image
created_by	UUID (FK)	Field Operator who scanned the batch

7.4 CustodyEvent

One record per custody transfer event in the chain of custody. Immutable once created.

Field	Type	Description
id	UUID (PK)	Unique event ID
batch_id	UUID (FK)	Linked mineral batch
event_type	ENUM	extraction / storage / dispatch / waypoint / receipt / export / rejection
from_party	TEXT	Name and role of transferring party
to_party	TEXT	Name and role of receiving party
gps_point	GEOMETRY (PostGIS POINT)	Location of transfer event
quantity_kg	DECIMAL	Quantity transferred in kg
authorised_by	UUID (FK)	User ID who authorised the transfer
timestamp	TIMESTAMP	UTC timestamp — set by server, not client
notes	TEXT (nullable)	Free text notes from the operator

7.5 SafetyIncident

Field	Type	Description
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id	UUID (PK)	Unique incident ID
org_id	UUID (FK)	Which organisation
site_id	UUID (FK)	Which site
zone_id	UUID (FK, nullable)	Linked mineral zone if applicable
incident_type	ENUM	gas_threshold / structural_instability / slope_failure / equipment / other
risk_score	INTEGER	Risk score at time of incident (0–100)
sensor_readings	JSONB	Snapshot of relevant sensor readings at incident time
gps_point	GEOMETRY (PostGIS POINT)	GPS location of incident
reported_by	UUID (FK)	User who reported (or system for automated alerts)
acknowledged_by	UUID (FK, nullable)	Safety Officer who acknowledged
acknowledged_at	TIMESTAMP (nullable)	When acknowledged
status	ENUM	open / acknowledged / resolved / escalated
created_at	TIMESTAMP	UTC timestamp

7.6 MLModelVersion

Field	Type	Description
id	UUID (PK)	Unique model version ID
version_number	TEXT	Semantic version string e.g. "2.1.3"
training_dataset_ref	TEXT	Reference to training dataset (S3 path or dataset ID)
accuracy_overall	DECIMAL	Overall accuracy on validation set (0–1)
accuracy_per_mineral	JSONB	Per-mineral-class precision, recall, F1 as JSON object
trained_by	UUID (FK)	AI Engineer who deployed the model
trained_at	TIMESTAMP	UTC time of training completion
deployed_at	TIMESTAMP (nullable)	UTC time deployed to production
deprecated_at	TIMESTAMP (nullable)	UTC time deprecated (never deleted)
is_active	BOOLEAN	Only one model version is active at a time per org
notes	TEXT (nullable)	Training notes and known limitations

8. Testing Approach & Go-Live Checklist

8.1 Testing Approach

Test Type	Description	When	Who
Unit Tests	Test one function or algorithm in isolation — especially classification algorithms, band ratio calculations, and data parsers	Written alongside every feature	Engineer who wrote it
API Integration Tests	Test complete API endpoints end-to-end with a real test database	After each module's backend tasks complete	Backend engineers
Geospatial Validation Tests	Verify that PostGIS queries return correct spatial results; test coordinate projection accuracy	After each GIS feature	Backend + GIS engineer
ML Accuracy Tests	Run classification model against held-out validation dataset; fail if accuracy drops below threshold	After every model retrain	AI engineers
Frontend E2E Tests	Automate real browser user flows (Playwright): upload, view map, generate report, tag batch	As each UI screen is completed	Frontend engineers
Offline Sync Tests	Test Field App offline queue, sync on reconnection, and conflict resolution	During Phase 2	Mobile + Backend engineers
Full System Tests	All three portals working together end-to-end on a simulated real site	Phase 4 and Phase 5	Full engineering team
Load Tests	Concurrent user load matching NFR targets; P95 < 500ms (k6 or Locust)	Before each phase deployment	Backend engineers
Security Penetration Test	External independent pen test: mandatory before Phase 4 deployment	Phase 4 pre-deployment	External security firm
UAT	Real mine operators on real or simulated data under MDMIS supervision	Phases 4–5	Operations team + end users

8.2 Go-Live Checklist (Phase 1 — Foundation)

Every item below must be verified and signed off before Phase 1 goes live. No exceptions.

#	Item	Verified By
1	All Phase 1 features deployed and working on production environment	All engineers
2	Zero mock API calls or placeholder data in any production build — confirmed by code scan	Lead engineer
3	Load test: 50 concurrent users; P95 API response < 500ms	Backend engineers
4	Security: no critical or high findings from internal review (external pen test at Phase 4)	Lead engineer + CTO

5	Sensor ingestion: all 7 sensor formats validated with real sample files from the client's hardware	GIS engineer + Backend
6	Multi-tenant isolation confirmed: verified that org A cannot access org B data via any API endpoint	Backend engineers
7	MFA working for all admin and compliance roles	Backend + QA
8	File upload: 1 GB GeoTIFF successfully uploaded, pre-processed, and stored within 5 minutes	Backend + GIS
9	Malware scanning (ClamAV) active and rejecting test malicious files	Backend
10	Email alerts (SendGrid) firing correctly for ingestion complete, safety alerts	Backend
11	SMS alerts (Africa's Talking) delivering for safety threshold breaches	Backend
12	Backup and restore tested: restore from backup completes within 4 hours	Backend engineers
13	Field App offline mode: queue data while offline, sync within 2 minutes on reconnection	Mobile + Backend
14	Map layers loading within 3 seconds for a site with 50+ scan sessions	Frontend + Backend
15	RBAC: each role confirmed to access only what is permitted — all 10 roles tested	QA engineers
16	Safety incident log: create, acknowledge, resolve flow confirmed end-to-end	QA + Operations
17	Error tracking (Sentry) active and alerting on production errors	Backend + CTO
18	Zero secrets (API keys, passwords) in code or version control — confirmed by git-secrets scan	CTO
19	User onboarding documentation and field operator training guide published	MDMIS Operations team
20	At least 1 real client organisation onboarded with real site data on staging	MDMIS Operations + Client

9. Phased Delivery Plan

Phase	Scope & Theme	Key Deliverables
Phase 1 — Foundation (Weeks 1–8)	Working data pipeline, authentication, and basic map display. The minimum system a real mine site can use.	Sensor data ingestion (all 7 formats). User auth with RBAC and MFA. Multi-site management. Basic mineral map display. Safety incident logging. Email + SMS alerts. Field App (online only). Multi-tenant isolation.
Phase 2 — Intelligence (Weeks 9–16)	AI classification engine live with real accuracy baseline. Offline Field App. Lab sample management.	ML mineral classifier (≥85% accuracy). Confidence scoring and anomaly flagging. Multi-layer map views. Lab sample database and model validation. Offline Field App with sync. Historical geological survey import. Resource estimation.
Phase 3 — Operations (Weeks 17–24)	Full mining operations management — traceability, grading, inventory, mine planning.	Mineral traceability (chain-of-custody, QR codes). Grading module with live commodity prices. Production and inventory ledger. Transportation tracking. Mine planning and excavation recommendations. Cross-site portfolio dashboard.
Phase 4 — Compliance (Weeks 25–30)	Regulatory readiness. OECD reports. RMB integration. External pen test.	OECD due diligence report generation. RMB formatted submission reports. 7-year immutable audit trail. Digital report signatures. Compliance Manager portal features. External penetration test completed and findings resolved.
Phase 5 — Advanced (Weeks 31–40)	Full production system. 3D visualisation. ERP integration. Public API.	3D subsurface visualisation (WebGL). Live commodity price integration. ERP/inventory system integration via REST API. Public API and webhooks for client integrations. SOC 2 Type II preparation. Advanced analytics and dashboards.

10. Data Requirements

10.1 Data Versioning

REQ-DV-001 [Phase 1]	Every dataset (scan session, model run, report) MUST have an immutable version record. A new version is created on any change — previous versions are preserved indefinitely.
REQ-DV-002 [Phase 2]	When new sensor data is added to an existing site, a new version of the mineral map MUST be generated automatically. Prior map versions remain accessible for comparison.
REQ-DV-003 [Phase 2]	Model versions MUST be tagged to every classification result. Users MUST be able to reprocess a scan session with a different model version for comparison.

10.2 Data Ownership & Retention

REQ-DO-001 [Phase 1]	Geological scan data, mineral maps, and resource estimates are the exclusive property of the client organisation. MDMIS has no right to share, sell, or use client data for any purpose other than operating the service for that client.
REQ-DO-002 [Phase 4]	All data MUST be retained for a minimum of 7 years to comply with Rwanda mining regulatory requirements and OECD due diligence standards.
REQ-DO-003 [Phase 5]	On client account termination, all data MUST be exportable in full in standard formats (GeoTIFF, GeoJSON, CSV, PDF). After confirmed export, all data MUST be securely deleted from all storage systems within 30 days.

11. Business Case Summary

This section summarises the financial and strategic justification for MDMIS. Full detail is in the separate Section 11 Business Case & Methodology document.

11.1 The Market Gap

KEY INSIGHT	Every system operating in this market today (iTSCi, Farmonaut, RMB databases) starts after the mineral is already dug up. MDMIS starts before the first shovel hits the ground — it is a different product category, not a competitor to what exists.
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11.2 Stakeholder Financial Impact

Stakeholder	Without MDMIS	With MDMIS	Est. Annual Benefit
Mining Companies	High exploration costs, inaccurate drilling	AI-assisted targeting, optimised extraction	\$100,000 – \$500,000
Mine Owners	Lower mineral recovery rates	Improved resource estimation and recovery	10–30% revenue increase
Government Agencies	Tax leakage and illegal mining	Enhanced traceability and monitoring	5–20% royalty increase
Mineral Exporters	Delayed compliance processes	Automated compliance reporting	\$10,000 – \$100,000 savings
Investors	High investment uncertainty	Reliable reserve and risk analysis	20–40% risk reduction
Safety Officers	Reactive, post-incident response	Predictive risk scoring and real-time alerts	>\$50,000 per incident prevented

11.3 Example Impact — Medium Coltan Mine

Metric	Without MDMIS → With MDMIS
Exploration Costs	\$200,000 → \$140,000 (30% reduction)
Mineral Recovery Revenue	\$500,000 → \$600,000 (20% increase)
Losses / Theft	\$40,000 → \$5,000 (87% reduction via traceability)
Compliance Costs	\$15,000 → \$5,000 (automated reporting)
Annual Profit	\$500,000 → \$715,000 (+\$215,000, +43%)

12. Open Items — Decisions Required Before Development

The following questions must be resolved with the client or stakeholders before the relevant engineering work can begin. Engineering should not guess on these items — flag and wait for sign-off.

#	Question	Why It Matters	Blocks
OQ-01	Which specific drone and sensor hardware will the client deploy? (Model numbers and firmware versions)	Determines exact input format parsers needed in Phase 1. Wrong assumption = broken ingestion.	Phase 1
OQ-02	Is cloud hosting acceptable or must the system run on-premise due to data security requirements at mine sites?	Affects all infrastructure design, offline capability scope, and network architecture.	Phase 1
OQ-03	What is the minimum acceptable ML classification accuracy? Client must define this formally in writing.	Sets the model acceptance threshold and determines how much training data must be collected before Phase 2 launch.	Phase 2
OQ-04	Which commodity price data provider will be used? Public API (LME) vs. paid market data feed?	Affects value estimation feature accuracy and integration cost.	Phase 3
OQ-05	Does the client have existing geological survey datasets to import as baseline reference data?	If yes, their format must be determined in Phase 1 planning. If no, baseline comparison features are limited.	Phase 2
OQ-06	What is the current RMB report format mandated? A sample template must be provided.	Cannot build the RMB submission module without the actual template.	Phase 4
OQ-07	What level of transportation tracking is required — vehicle GPS tracking or shipment manifests only?	GPS tracking requires hardware integration (tracker devices). Manifests only is software-only.	Phase 3
OQ-08	Which lab information system does the client's ground truth lab use? Is an API available?	Determines whether lab results can be auto-ingested or must be manually uploaded as CSV.	Phase 2
OQ-09	Are there specific artisanal mining cooperative workflows that differ from formal mining operations?	ASM workflows often lack structured inventory tracking — may require simplified Field App screens.	Phase 1

13. Document Control

Field	Value
Document Title	Mineral Detection & Mining Intelligence System — Software Requirements Specification
Version	2.0
Date	June 2026
Status	Draft — Pending CEO & CTO Sign-off
Prepared By	MDMIS Product & Engineering Team
Target Region	Rwanda / Eastern DRC (Goma Corridor)
Classification	Confidential — Development Use Only
Next Review	After stakeholder sign-off on Open Items (Section 12)
Version History	v1.0 (May 2026): Initial draft. v2.0 (June 2026): Full expansion with REQ codes, role matrix, database schemas, glossary, workflow tables, SLA specs, go-live checklist, and open items.

SIGN-OFF REQUIRED	<i>This document requires formal sign-off from the CEO and CTO before any development team begins Phase 1 work. Open Items in Section 12 must be resolved and recorded in the sign-off meeting minutes.</i>
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